Haoyang Zhang

Electricity Markets | Optimization | Flexibility

Belgium | zhanghaoyang1993.github.io | haoyang-zhang-272397138 | zhanghaoyang1993

Professional profile

Energy market modeller with a PhD from Eindhoven University of Technology. Research spans electricity market design, optimization and multi-agent reinforcement learning. Current work at VITO / EnergyVille focuses on Core flow-based day-ahead market reconstruction and residential flexibility. Work connects market rules, strategic participant behaviour and physical network constraints, from wholesale market integration to local flexibility procurement.

Experience

Energy Market Modelling Expert | VITO / EnergyVille
November 2025 - present · Belgium

European electricity market reconstruction and residential flexibility modelling.

Visiting Researcher | Empa
March - July 2025 · Switzerland

- Investigated bidding across sequential local markets using hierarchical multi-agent reinforcement learning; co-authored a research preprint with Empa and TU/e collaborators.

Doctoral Researcher | Eindhoven University of Technology
October 2021 - October 2025; PhD awarded June 2026 · Netherlands

- Developed a simulation framework for local electricity and flexibility markets to study strategic bidding under distribution network constraints.
- Compared learned bidding behaviour with truthful-bidding benchmarks and assessed multi-agent learning reliability using a no-regret measure.
- Evaluated pricing mechanisms and market power detection methods in simulated local markets.

Associate Researcher | China Southern Power Grid Energy Development Research Institute
2019 - 2021 · China

Energy policy and electricity market analysis.

Current projects

CROCODILE | VITO / EnergyVille

- Reconstruct the Core flow-based day-ahead electricity market using public JAO/NEMO data to support historical market analysis.
- Use historical Core market reconstruction to support analysis of offshore electricity market design and integration.

EU-DREAM | VITO / EnergyVille

- Model residential flexibility from heat pumps and EV charging under dynamic electricity tariffs and capacity-based network charging.

Selected research & personal project

MEGAMIND

Eindhoven University of Technology | PhD research completed in 2026

- Designed a hierarchical local market framework using multi-cut Benders decomposition; published the method in Applied Energy (2024).
- Assessed multi-agent reinforcement learning reliability for hybrid local market simulation; published in Applied Energy (2025).

EuroPowerQuantLab

Independent personal project | Under development

- Developing an independent Python-based research platform for European electricity market data, forecasting research and historical strategy evaluation; under development.

Education

PhD in Electrical Engineering | Eindhoven University of Technology | 2026

MSc in Electric Power Engineering | KTH Royal Institute of Technology | 2019
Exchange and master's thesis research at ETH Zürich Power Systems Laboratory, supervised by Gabriela Hug and Thierry Zufferey; distribution grid state estimation.

BSc in Electrical Engineering and Automation | South China University of Technology | 2016

Technical skills

Programming: Python; MATLAB; SQL; Gurobi; PyTorch

Quantitative methods: Mathematical optimization; multi-cut Benders decomposition; reinforcement learning; multi-agent simulation

Electricity markets: Historical market reconstruction; strategic bidding simulation; local electricity and flexibility markets

Networks & flexibility: Flow-based market coupling; CNEC/PTDF/RAM; congestion management; heat pumps; EV charging; tariff modelling

Selected publications

Haoyang Zhang, Sen Zhan, Koen Kok, Nikolaos G. Paterakis. **Establishing a hierarchical local market structure using multi-cut Benders decomposition**. *Applied Energy*, 2024. DOI

Haoyang Zhang, Dawei Qiu, Koen Kok, Nikolaos G. Paterakis. **Reliability assessment of multi-agent reinforcement learning algorithms for hybrid local electricity market simulation**. *Applied Energy*, 2025. DOI

Xin Li, Sen Zhan, Haoyang Zhang, Koen Kok, Nikolaos G. Paterakis. **A privacy-preserving distributed congestion management framework via implicit differentiation bilevel optimization**. *Electric Power Systems Research*, 2027 (online 2026; volume 2027). DOI

Awards & professional development

2025: EEM Best Student Paper Award | European Energy Market conference

2025: NCCR Automation Fellowship | NCCR Automation / Empa research visit

November 2024: The Physical and Financial Power | Nord Pool Academy